

Engineering Thesis

Can Demographic Structure of GP Practice Populations Improve the Explanation of Variation in Hospital Service Use in Scotland?

Assignment 10 — Methodology and Technical Chapters

Chapter 3: Methodology

3.1 Research Framework and Experimental Design

This research investigates whether the demographic structure of General Practitioner (GP) practice populations provides explanatory power for variation in hospital service use across practices in Scotland, beyond what is captured by practice size and GP workforce availability alone. The formal research question is: Can demographic structure of GP practice populations improve the explanation of variation in hospital service use in Scotland compared with a baseline model based only on practice size and GP availability?

The experimental design follows a comparative modelling approach structured around two feature sets that reflect a progressive expansion of explanatory variables. The first set, designated F0, constitutes the baseline specification and contains only practice population size (log-transformed) and GP availability. The second set, designated F1, extends F0 by incorporating demographic composition variables — specifically the proportion of the population aged 65 or older, the proportion female, and a deprivation index — together with regional categorical controls for Health Board (HB) and Health and Social Care Partnership (HSCP) area.

The choice of this two-set structure is theoretically motivated. Population ageing and deprivation are well-established drivers of unplanned hospital admissions and emergency service demand in the health economics literature. If demographic composition exerts a systematic influence on hospital utilisation above and beyond practice headcount and staffing, that influence should manifest as a measurable improvement in predictive performance when moving from F0 to F1.

The comparison logic across models is designed to isolate the contribution of feature set expansion from the contribution of model complexity. Accordingly, each feature set is evaluated across multiple model families: a mean predictor (Dummy), ordinary least squares

(OLS), ridge regression, and a decision tree. A random forest is additionally included in the final experiment stage as a nonlinear benchmark. The primary evaluation criterion is holdout Root Mean Squared Error (RMSE), supported by the coefficient of determination (R^2) and Mean Absolute Error (MAE) as secondary measures.

The evaluation strategy is deliberately conservative. All preprocessing transformations and hyperparameter selection are performed exclusively on the training partition, with the holdout set withheld entirely until final model assessment. Cross-validation is conducted at the group level, using GP practice as the grouping unit, to prevent information leakage across temporal observations from the same practice. This design choice is necessitated by the panel structure of the dataset, in which each practice contributes multiple monthly records.

3.2 Dataset Description

The analytical dataset is constructed by merging two source tables derived from Public Health Scotland administrative records. The first table contains GP practice population demographics, disaggregated by sex and age band at the monthly level. The second table contains practice-level supporting variables, including a measure of GP workforce availability, a deprivation index, and the target variable: hospital admissions per 1,000 registered patients.

The unit of observation is a GP practice in a given calendar month, identified by the combination of PracticeCode and Date. Practices are nested within Health Board regions and Health and Social Care Partnership areas, which introduce a regional grouping structure that the categorical controls in F1 are intended to capture. The dataset has a balanced panel character: each practice contributes observations across consecutive months, though the effective sample used in the reproducibility demonstration consists of a synthetic four-practice, three-month bundle prepared specifically to confirm that the full pipeline is runnable under controlled conditions.

The demographic table records population counts across eight age bands (0–4, 5–14, 15–24, 25–44, 45–64, 65–74, 75–84, 85 and over) separately for male, female, and all-persons aggregates. The analytical pipeline retains only the all-persons rows for age-share derivation and merges the female-persons count to compute the sex composition ratio. Records in which the total registered population is zero or missing are excluded prior to any feature computation.

The target variable, `hospital_use_per_1000`, is sourced from the supporting input table and represents the monthly rate of hospital admissions or contacts per 1,000 registered patients. This rate-based formulation controls for the confounding effect of practice size on the raw admission count and is consistent with standard practice in health services research. The deprivation index is a composite measure on a 0–100 scale, with higher values indicating greater material deprivation. The GP availability variable is a continuous positive ratio reflecting the number of whole-time-equivalent GPs relative to the registered population.

The principal data quality constraint is the reliance on a synthetic sample bundle for the demonstrated experimental runs. While the pipeline architecture, preprocessing logic, and modelling framework are fully developed and validated against this bundle, the resulting performance metrics are methodological proofs-of-concept rather than substantive empirical estimates. The same pipeline is intended to be rerun on the full administrative dataset without modification to the source code or configuration structure.

Table 3.1 — Dataset characteristics summary

Attribute	Value
Observation unit	GP practice × calendar month
Identifier columns	PracticeCode, Date
Regional groupings	Health Board (HB), HSCP area
Age bands recorded	8 (0–4 through 85+), all-persons aggregate
Target variable	hospital use per 1000 (admissions per 1,000 patients)
Supporting variables	gp availability, deprivation index
Train / test split	80% training practices / 20% holdout practices
Cross-validation	5-fold, group-aware (practice-level)
Random seed	42
Demonstrated sample	4 practices, 3 months (synthetic bundle)

3.3 Preprocessing and Feature Engineering

The preprocessing pipeline is designed to be fitted exclusively on training data and applied identically to held-out data, with no information flowing from the test partition into any transformation parameter. This constraint is implemented by computing all scaling statistics and encoding mappings from the training subset before projection onto the validation or holdout sets.

The first preprocessing step is the reconstruction of sex composition from the raw demographics table. The all-persons records are retained as the base, and the female-persons count is merged using the practice–date key. Where female headcount is absent, it is imputed as 50% of the total registered population. This imputation is applied before any split, as it is a structural reconstruction step rather than a learned transformation.

Three demographic ratio features are derived directly from the age-band counts and the merged female population. The share of the population aged 0–14 is computed as the sum of the 0–4 and 5–14 bands divided by the all-ages total. The share aged 65 or older sums the 65–74, 75–84, and 85+ bands in the same manner. The share female divides the female headcount by the all-ages total. These ratios are unit-free and lie in the interval $[0, 1]$, removing the scale dependency of the raw counts.

Practice size is represented through a log-transformed total population count, computed as $\log(1 + \text{AllAges})$. The logarithmic transformation is applied because the distribution of registered practice populations is right-skewed across Scottish GP practices, with a small number of large urban practices spanning an order of magnitude above the median. The log-

transformation compresses this range, reducing the leverage of large practices in linear model fitting and making the coefficient interpretable as an elasticity.

Numeric features entering the model are standardised using the mean and standard deviation computed from the training set. This step is necessary for ridge regression, where the L2 penalty treats all coefficients on a common scale; without standardisation, the regularisation effect would be disproportionately applied to features with larger raw variance. The standardisation parameters are stored from the training design matrix and re-applied without recomputation during holdout projection.

Categorical variables, Health Board and HSCP area, are encoded using one-hot encoding with the category levels fixed to those present in the training data. Unseen categories at test time are mapped to an all-zero vector. Mode imputation is applied to any missing categorical values prior to encoding, using the modal category from the training fold. Missing numeric values are imputed using the training-set median.

Feature selection is not applied in this study. The rationale is that the research question is already operationalised through a pre-defined contrast between two feature sets rather than through a data-driven selection procedure. Introducing selection would conflate the contribution of demographic variables with the selection criterion and would complicate the interpretability of the F0-to-F1 comparison.

Table 3.2 — Feature sets used in the experiments

Feature	Set	Type	Derivation / Source
log all ages	F0, F1	Numeric	$\log(1 + \text{AllAges})$; practice population
gp availability	F0, F1	Numeric	Supporting inputs; GP workforce ratio
share age 65 plus	F1	Numeric	$\Sigma(\text{Ages65-74, 75-84, 85+}) / \text{AllAges}$
share female	F1	Numeric	female population / AllAges
deprivation index	F1	Numeric	Supporting inputs; 0–100 composite index
HB	F1	Categorical (OHE)	Health Board region
HSCP	F1	Categorical (OHE)	Health and Social Care Partnership area

3.4 Evaluation Strategy

The evaluation strategy is structured around two complementary assessment mechanisms: a practice-level holdout partition and a group-aware cross-validation procedure. The panel nature of the dataset requires that temporal replicates from the same practice do not appear simultaneously in training and evaluation sets, as this would constitute a form of data leakage that artificially inflates performance estimates.

The holdout partition assigns complete practices to either the training or the test set. The split is determined using the global random seed (42) applied to the sorted list of unique practice codes, with 20% of practices allocated to the holdout. All temporal records for a given practice remain in a single partition. The holdout set is withheld from all preprocessing parameter estimation, hyperparameter selection, and model fitting procedures.

Cross-validation is performed on the training partition only. Folds are constructed at the practice level: each fold designates one or more complete practices as the validation set, with the remaining practices serving as the training fold. With the demonstrated four-practice sample, this produces three folds. On the full dataset, five group-aware folds are used. Preprocessing parameters are recomputed independently within each fold.

The primary optimisation criterion for hyperparameter selection is cross-validated RMSE. Holdout metrics are computed once, after hyperparameter selection is complete, and serve as the final reported performance. The three reported metrics — R^2 , RMSE, and MAE — provide complementary perspectives: R^2 expresses the proportion of variance explained, RMSE penalises large errors quadratically, and MAE provides a robust absolute-error summary.

Chapter 4: Technical Implementation

4.1 Pipeline Architecture

The technical implementation is structured as a modular pipeline with a clear separation between data ingestion, preprocessing, model training, and result tracking. The pipeline entry point accepts a YAML configuration file that specifies all experiment-level parameters, including feature sets, model type, hyperparameters, file paths, and the global random seed. This configuration-driven approach ensures that different experimental conditions are isolated at the configuration layer rather than through code duplication.

The preprocessing stage reads both raw input tables, validates column presence, derives the engineered features, and writes a processed analytical dataset together with a manifest file recording the SHA-256 hashes of the input files, row counts, column lists, and the timestamp of the processing run. The manifest enables downstream verification that a given trained model corresponds to a specific version of the processed data.

The training stage constructs the design matrix from the processed dataset, applies all data-dependent transformations, fits the selected model, and computes both holdout and cross-validation metrics. Outputs are written to a run directory whose name is derived deterministically from the experiment name, configuration hash, and random seed. This ensures that reruns of identical configurations produce identically named output directories, which simplifies reproducibility verification.

4.2 Baseline Models

Four baseline models are implemented to establish the reference performance landscape prior to primary model development.

The Dummy mean predictor (F0) provides the unconditional reference by predicting the training-set target mean for all observations. This model has no learnable parameters

conditioned on input features and therefore represents the minimum performance that any informative model must exceed.

The OLS baseline (F0) applies ordinary least squares regression to the two-feature baseline set. No regularisation is applied. The OLS solution is obtained via the closed-form normal equations, and the model is evaluated on the holdout partition without refitting. The OLS F0 result serves as the primary reference point for quantifying the incremental contribution of demographic variables, because it isolates the maximum-likelihood linear relationship between practice size, GP availability, and the target.

The Ridge F1 baseline (Assignment 7) applies L2-regularised linear regression to the extended feature set with a limited alpha grid of {0.1, 1.0, 10.0}, selected by cross-validated RMSE. This model tests whether the regularised linear projection of demographic variables onto the target improves over the unregularised OLS baseline. The inclusion of categorical encodings for HB and HSCP in F1 increases the effective dimensionality relative to F0, making regularisation appropriate.

The Decision Tree benchmark (F1) applies a classification and regression tree with a maximum depth of three and a minimum leaf size of two. No hyperparameter tuning is performed for the tree, as the assignment scope specifies it as a structural complexity benchmark rather than an optimised model.

4.3 Primary Model: Ridge Regression F1

The primary model is L2-regularised linear regression (ridge regression) applied to the F1 feature set. Ridge regression is chosen as the primary model for three reasons: its coefficients remain analytically interpretable after standardisation, the L2 penalty provides stable estimation when features are correlated (as may be expected among age shares and the deprivation index), and it extends naturally to the multi-category encoding required by the regional categorical variables.

The regularisation strength parameter alpha is selected from an expanded grid of seven values spanning seven orders of magnitude: {0.001, 0.01, 0.1, 1.0, 10.0, 100.0, 1000.0}. Selection is based on the cross-validated mean RMSE across the group-aware folds constructed from the training partition. The optimal value identified under this criterion is $\alpha = 0.01$, which achieves a CV mean RMSE of 4.84 on the demonstrated sample. The expansion of the alpha grid relative to Assignment 7 is motivated by the observation that the optimal Assignment 7 value ($\alpha = 0.1$) lay at the boundary of the earlier grid, warranting investigation of lower regularisation strengths.

The model is fit on the full training partition using the selected alpha value and then evaluated once on the holdout set. The closed-form ridge solution is computed as $\beta = (Z^T Z + \alpha I^*)^{-1} Z^T y$, where Z is the augmented design matrix (with an intercept column), I^* is the identity matrix

with the intercept diagonal entry set to zero to leave the intercept unpenalised, and α is the selected regularisation parameter.

The resulting coefficient vector is directly interpretable in terms of the standardised feature scale: positive coefficients indicate a positive association between the feature and hospital utilisation rate, negative coefficients the reverse. The intercept represents the model-predicted hospital use rate for a practice at the mean of all continuous features and the reference categories for HB and HSCP.

4.4 Nonlinear Benchmark: Random Forest

A random forest regressor is included in the final experimental stage as a nonlinear benchmark. Its inclusion serves to assess whether nonlinear interactions among demographic features and regional controls provide predictive gains beyond what the ridge regression can capture through linear projection.

The forest is configured with a grid of six hyperparameter combinations, varying the number of estimators across $\{50, 100, 200\}$, the maximum tree depth across $\{2, 3\}$, and the minimum samples per leaf across $\{2, 4\}$. Selection is performed by group-aware cross-validated RMSE, yielding the configuration $n_estimators = 200$, $max_depth = 3$, $min_samples_leaf = 2$ as the cross-validation optimum on the demonstrated sample.

The random forest results on the demonstrated sample are not interpretable as substantive performance estimates. With only three training practices contributing nine rows, the forest overfits severely: the holdout R^2 is -297 , compared with $+0.47$ for the primary ridge model. This divergence is attributable to the extreme data scarcity of the synthetic demonstration bundle. On the full administrative dataset, the relative performance of the two model families is expected to be more informative. The forest's feature importance rankings, however, are included for interpretability purposes and are consistent with domain expectations: deprivation index, share female, and share aged 65+ are the top-ranked predictors.

4.5 Comparative Results Summary

Table 4.1 consolidates the holdout and cross-validation performance metrics across all models included in the experimental sequence. Results are reported for both the Assignment 7 baseline experiment and the Assignment 8 extended experiment. The OLS F0 model serves as the reference for computing relative RMSE improvement.

Table 4.1 — Consolidated model comparison (demonstrated sample)

Model	Features	Holdout R^2	Holdout RMSE	Holdout MAE	CV RMSE (mean)	RMSE improv. vs OLS F0
Dummy mean (A7)	F0	-818.57	52.63	52.60	45.87	-582.0%
OLS F0 (A7)	F0	-16.62	7.72	7.54	43.96	reference
Ridge F1 $\alpha=0.1$ (A7)	F1	-18.36	8.09	8.08	8.31	-4.8%
Decision Tree F1 (A7)	F1	-438.54	38.54	38.50	57.63	-399.5%

Ridge F1 $\alpha=0.01$ (A8)	F1	+0.469	1.34	1.34	4.84	+82.6%
Random Forest F1 (A8)	F1	-297.00	31.74	31.68	34.17	-311.3%

The results demonstrate that the alpha-expanded Ridge F1 model achieves a substantial improvement in holdout RMSE (82.6%) and a positive R^2 value (+0.47) compared with the OLS F0 baseline. This confirms that the demographic feature set, combined with appropriate regularisation, provides measurable explanatory gain over the practice-size-and-availability baseline. The Assignment 7 Ridge F1 configuration did not reproduce this gain because the alpha grid was insufficient to identify the optimal regularisation strength. The expanded search in Assignment 8 corrects this limitation.

4.6 Experimental Environment

All experiments are executed on a single machine running Windows 11. The hardware configuration consists of an AMD Ryzen AI 9 HX 370 processor with integrated AMD Radeon 890M graphics and 30.62 GB of RAM. No GPU acceleration is required or used; all computations are performed on CPU. The pipeline is designed to be runnable on any modern workstation and does not depend on specialised hardware.

The software environment for the core pipeline (Assignment 4) is pinned to Python 3.11.9. The modelling notebooks for Assignments 7 and 8 are executed under Python 3.14.4. The primary scientific computing libraries are NumPy, pandas, and scikit-learn; matplotlib is used for diagnostic visualisations; python-docx is used for report generation. The full dependency list is recorded in requirements.txt and environment.yml at the repository root of the Assignment 4 project directory.

Reproducibility is enforced through three mechanisms. First, the global random seed (42) is set before any stochastic operation, including train–test splitting, cross-validation fold assignment, and random forest bootstrap sampling. Second, experiment configurations are defined in YAML files and are not modified during execution; the merged configuration snapshot is written alongside each run's outputs. Third, the SHA-256 hash of each input file is recorded in the preprocessing manifest, ensuring that a change in the input data is detectable before any model is trained.

Table 4.2 — Software environment summary

Component	Version / Detail
Python (pipeline)	3.11.9
Python (notebooks A7–A8)	3.14.4
Operating system	Windows 11 (10.0.26200)
Processor	AMD Ryzen AI 9 HX 370 w/ Radeon 890M
RAM	30.62 GB
NumPy	pinned in requirements.txt
pandas	pinned in requirements.txt
scikit-learn	pinned in requirements.txt
matplotlib	pinned in requirements.txt

python-docx	pinned in requirements.txt
Global random seed	42
Experiment configuration	YAML (per-experiment, hash-tracked)

Chapter 5: Discussion of Methodological Decisions

Several methodological decisions require explicit justification to clarify the scope and limitations of the findings.

The choice of ridge regression as the primary model reflects a deliberate preference for interpretability and stability over predictive flexibility. Given the policy-relevant context of the research — understanding which demographic factors systematically relate to hospital demand — a model with recoverable and signed coefficients is analytically preferable to a black-box ensemble. The random forest serves as a validity check on the linearity assumption rather than as a candidate for deployment.

The use of a practice-level group split rather than a time-based split reflects the priority of measuring generalisation across practices rather than across time. A temporal split would prevent data leakage in a forecasting context but would introduce a temporal confound that is not the primary concern of this cross-sectional comparison. If the research question were reformulated as a forecasting task, the evaluation strategy would need to be revised accordingly.

The exclusion of interaction terms between demographic variables and regional indicators is a deliberate simplification. While such terms might capture heterogeneous demographic effects across Health Board regions, their inclusion would substantially increase the number of estimated parameters relative to the available sample size, exacerbating the overfitting risk already observed in the random forest results on the demonstration data.

The synthetic nature of the demonstration bundle constitutes the primary limitation of the results reported in this chapter. The dataset contains 12 observations from 4 practices across 3 months. Metrics derived from this bundle — particularly the holdout R^2 of +0.47 for the primary ridge model — should not be interpreted as estimates of the model's performance on the full administrative dataset. The demonstration bundle is designed solely to confirm that the pipeline is executable and that the experimental framework is correctly implemented. Substantive empirical conclusions will be drawn from the full dataset in subsequent work.